

NORYL GTXTM RESIN GTX989

REGION AMERICAS

DESCRIPTION

NORYL GTXTM 989 resin is a conductive, non-reinforced alloy of Polyphenylene Ether (PPE) + Polyamide (PA). This injection moldable grade is optimized for primer-less electrostatic painting. NORYL GTX989 resin exhibits high heat resistance, high impact resistance, and low coefficient of thermal expansion (CTE) of ~9. This material is an excellent candidate for automotive applications such as body panels, tank flaps, fenders, trunk lid, and exterior trim.

TYPICAL PROPERTY VALUES

Revision 20200610

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
MECHANICAL			
Tensile Stress, yield, 50 mm/min	60	MPa	ISO 527
Tensile Stress, break, 50 mm/min	55	MPa	ISO 527
Tensile Strain, break, 50 mm/min	40	%	ISO 527
Tensile Strain, yield, 50 mm/min	5	%	ISO 527
Tensile Modulus, 1 mm/min	2300	MPa	ISO 527
Tensile Stress, break, 5 mm/min	57	MPa	ISO 527
Tensile Strain, break, 5 mm/min	45	%	ISO 527
Tensile Modulus, 1 mm/min @ 120°C	2265	MPa	ISO 527
Flexural Stress, yield, 2 mm/min	90	MPa	ISO 178
Flexural Modulus, 2 mm/min	2300	MPa	ISO 178
Tensile Strain, yld, Type I, 50 mm/min	5	%	ASTM D 638
Tensile Modulus, 50 mm/min	2350	MPa	ASTM D 638
Tensile Stress, brk, Type I, 50 mm/min	60	MPa	ASTM D 638
Tensile Strain, brk, Type I, 50 mm/min	45	%	ASTM D 638
Tensile Stress, yld, Type I, 50 mm/min	65	MPa	ASTM D 638
Tensile Modulus, 5 mm/min	2276	MPa	ASTM D 638
Tensile Strain, brk, Type I, 5 mm/min	39	%	ASTM D 638
Tensile Stress, brk, Type I, 5 mm/min	55	MPa	ASTM D 638
Flexural Modulus, 1.3 mm/min, 50 mm span	2220	MPa	ASTM D 790
Flexural Stress, yld, 1.3 mm/min, 50 mm span	95	MPa	ASTM D 790
IMPACT			
Izod Impact, notched 80*10*4 +23°C	22	kJ/m ²	ISO 180/1A
Izod Impact, notched 80*10*4 -30°C	15	kJ/m ²	ISO 180/1A
Izod Impact, unnotched, 23°C	DB	J/m	ASTM D 4812
Izod Impact, notched, 23°C	251	J/m	ASTM D 256
Izod Impact, notched, -30°C	180	J/m	ASTM D 256
Charpy 23°C, V-notch Edgew 80*10*4 sp=62mm	25	kJ/m ²	ISO 179/1eA
Charpy 23°C, Unnotch Edgew 80*10*4 sp=62mm	DB	kJ/m ²	ISO 179/1eU
Charpy -30°C, V-notch Edgew 80*10*4 sp=62mm	15	kJ/m ²	ISO 179/1eA
Instrumented Impact Total Energy, 23°C	60	J	ASTM D 3763
THERMAL			
Vicat Softening Temp, Rate A/50	246	°C	ISO 306
Vicat Softening Temp, Rate B/50	166	°C	ISO 306

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
Vicat Softening Temp, Rate B/120	200	°C	ISO 306
Vicat Softening Temp, Rate B/50	246	°C	ASTM D 1525
HDT, 0.45 MPa, 3.2 mm, unannealed	190	°C	ASTM D 648
HDT, 1.82 MPa, 3.2mm, unannealed	133	°C	ASTM D 648
HDT/Af, 1.8 MPa Flatw 80*10*4 sp=64mm	123	°C	ISO 75 /Af
HDT/Bf, 0.45 MPa Flatw 80*10*4 sp=64mm	190	°C	ISO 75 /Bf
HDT/Be, 0.45MPa Edgew 120*10*4 sp=100mm	190	°C	ISO 75 /Be
CTE, -40°C to 60°C, flow	8.5E-05	1/°C	ASTM E 831
CTE, -40°C to 60°C, xflow	8.5E-05	1/°C	ASTM E 831
CTE, 23°C to 60°C, flow	9.3E-05	1/°C	ISO 11359-2
CTE, 23°C to 60°C, xflow	9.5E-05	1/°C	ISO 11359-2
CTE, 23°C to 80°C, flow	9.3E-05	1/°C	ASTM E 831
CTE, 23°C to 80°C, xflow	9.5E-05	1/°C	ASTM E 831
PHYSICAL			
Specific Gravity	1.08	-	ASTM D 792
Density	1.08	g/cm³	ISO 1183
Water Absorption, (23°C/saturated)	2.29	%	ISO 62-1
Moisture Absorption (23°C / 50% RH)	0.64	%	ISO 62
Moisture Absorption, (23°C/50% RH/24 hrs)	0.22	%	ASTM D 570
Mold Shrinkage, flow, 24 hrs	1.56	%	ISO 294
Mold Shrinkage, xflow, 24 hrs	1.69	%	ISO 294
Mold Shrinkage, flow, 3.2 mm	1.2 – 1.6	%	SABIC method
Mold Shrinkage, flow, 24 hrs	1.56	%	ASTM D 955
Mold Shrinkage, xflow, 24 hrs	1.69	%	ASTM D 955
Melt Volume Rate, MVR at 280°C/5.0 kg	17	cm³/10 min	ISO 1133
Melt Flow Rate, 280°C/5.0 kgf	16	g/10 min	ASTM D 1238
ELECTRICAL			
Volume Resistivity	1.E+03 – 1.E+04	Ohm-cm	SABIC method
INJECTION MOLDING			
Drying Temperature	100 – 120	°C	
Drying Time	2 – 3	hrs	
Maximum Moisture Content	0.07	%	
Minimum Moisture Content	0.02	%	
Melt Temperature	290 – 320	°C	
Nozzle Temperature	280 – 310	°C	
Front - Zone 3 Temperature	290 – 320	°C	
Middle - Zone 2 Temperature	280 – 300	°C	
Rear - Zone 1 Temperature	260 – 280	°C	
Hopper Temperature	60 – 80	°C	
Mold Temperature	100 – 120	°C	



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